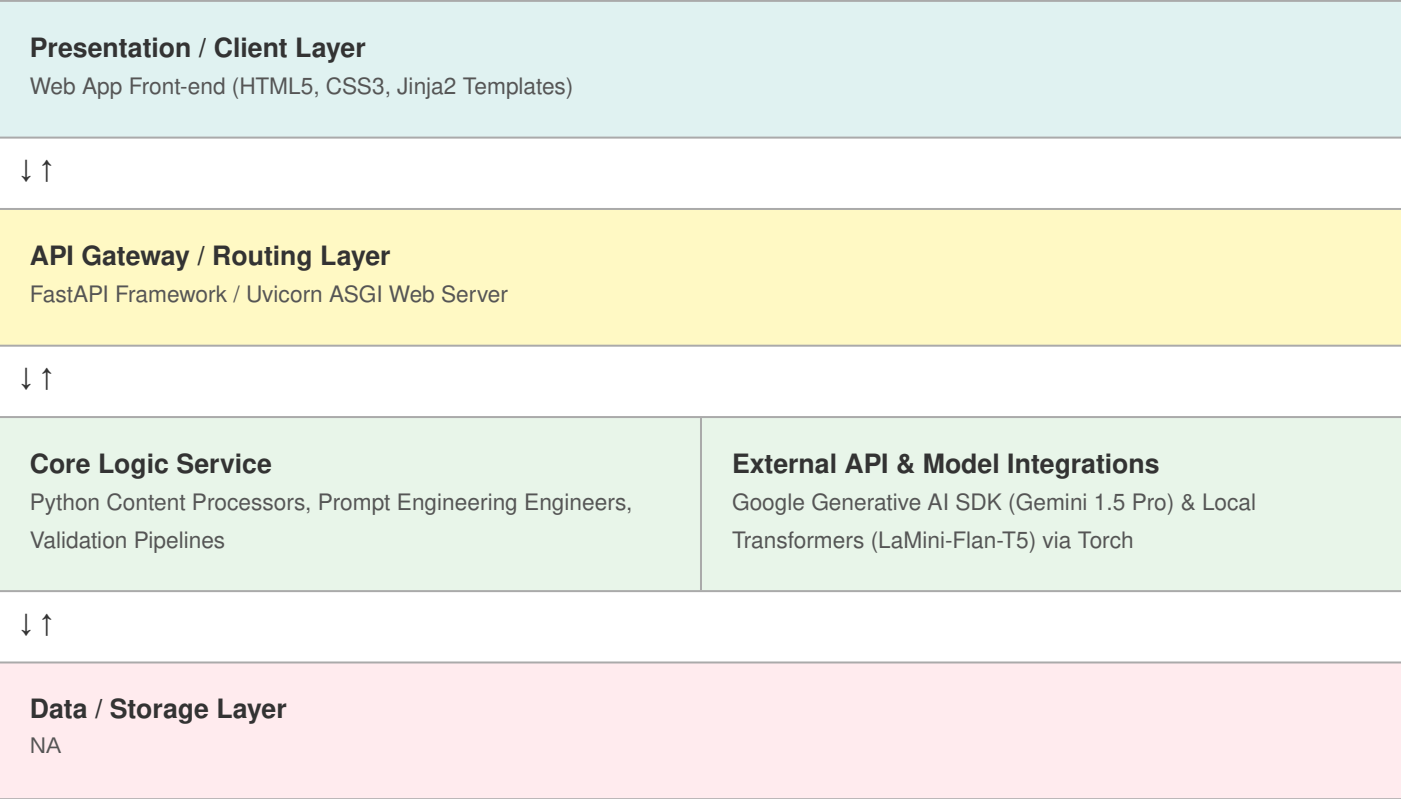


SOLUTION ARCHITECTURE & COMPONENTS

Solution Architecture Diagram Layout



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders interactive forms, handles input query submissions, displays real-time AI answers, summaries, and dynamically formats generated quizzes.	HTML5, CSS3, Jinja2 Templates
API Gateway	Acts as the core routing backend server, exposing fast REST endpoints for question handling, text summarization, and query execution requests.	FastAPI, Uvicorn
Microservices / AI Core	Manages model inference pipelines, prompt engineering logic, content post-processing, validation, and handles calls to heavy cloud or local networks.	Python, Google GenAI SDK, Transformers, Torch